

Note that the left-hand side is free of w_i because the two terms involving w_i cancel each other out. This is because the term $\bar{d}_i^T \bar{r}$ contributes $-w_i \bar{d}_i^T \bar{d}_i$, which cancels with $w_i \bar{d}_i^T \bar{d}_i$. Because of the fact that one of the sides does not depend on w_i , we obtain an update that yields the optimal value of w_i in a *single* iteration:

$$\bar{w}_i \leftarrow \bar{w}_i + \frac{\bar{d}_i^T \bar{r}}{\|\bar{d}_i\|^2} \quad (4.74)$$

In the above update, we have used the fact that $\bar{d}_i^T \bar{d}_i$ is the same as the squared norm of $\bar{d}_i$. It is common to standardize each column of the data matrix to zero mean and unit variance. In such a case, the value of $\|\bar{d}_i\|^2$ will be 1, and the update further simplifies to the following:

$$\bar{w}_i \leftarrow \bar{w}_i + \bar{d}_i^T \bar{r} \quad (4.75)$$

This update is extremely efficient. One full cycle of coordinate descent through all the variables requires asymptotically similar time as one full cycle of stochastic gradient descent through all the points. However, the number of cycles required by coordinate descent tends to be smaller than that in least-squares regression. Therefore, the coordinate-descent approach is more efficient. One can also derive a form of coordinate descent for regularized least-squares regression. We leave this problem as a practice exercise.

Problem 4.10.1 *Show that if Tikhonov regularization is used with parameter λ on least-squares regression, then the update of Equation 4.74 needs to be modified to the following:*

$$w_i \leftarrow \frac{w_i \|\bar{d}_i\|^2 + \bar{d}_i^T \bar{r}}{\|\bar{d}_i\|^2 + \lambda}$$

The simplification of optimization subproblems that are inherent in solving for one variable at a time (while keeping others fixed) is very significant in coordinate descent.

4.10.2 Block Coordinate Descent

Block coordinate descent generalizes coordinate descent by optimizing a *block* of variables at a time, rather than a single variable. Although each step in block coordinate descent is more expensive, fewer steps are required. An example of block coordinate descent is the *alternating least-squares method*, which is often used in matrix factorization (cf Section 8.3.2.3 of Chapter 8). Block coordinate descent is often used in multi-convex problems where the objective function is non-convex, but each block of variables can be used to create a convex subproblem. Alternatively, each block admits to easy optimization, even when some of the variables are discrete. It is sometimes also easy to handle constrained optimization problems with coordinate descent, because the constraints tend to simplify themselves, when one is considering only a few carefully chosen variables. A specific example of this type of setting is the k -means algorithm.

4.10.3 K-Means as Block Coordinate Descent

The k -means algorithm is a good example of how choosing specific blocks of variables carefully allows good alternating minimization over different blocks of variables. One often views k -means as a simple heuristic method, although the reality is that it is fundamentally rooted in important ideas from coordinate descent.

It is assumed that there are a total of n data points denoted by the d -dimensional row vectors $\bar{X}_1 \dots \bar{X}_n$. The k -means algorithm creates k *prototypes*, which are denoted by $\bar{z}_1 \dots \bar{z}_k$, so that the sum of squared distances of the data points from their nearest prototypes is as small as possible. Let y_{ij} be a 0-1 indicator of whether point i gets assigned to cluster j . Each point gets assigned to only a single cluster, and therefore we have $\sum_j y_{ij} = 1$. One can therefore, formulate the k -means problem as a *mixed integer program* over the *real-valued* d -dimensional prototype row vectors $\bar{z}_1 \dots \bar{z}_k$ and the matrix $Y = [y_{ij}]_{n \times k}$ of *discrete* assignment variables:

$$\begin{aligned} & \text{Minimize } \sum_{j=1}^k \underbrace{\sum_{i=1}^n y_{ij} \|\bar{X}_i - \bar{z}_j\|^2}_{O_j} \\ & \text{subject to:} \\ & \sum_{j=1}^k y_{ij} = 1 \\ & y_{ij} \in \{0, 1\} \end{aligned}$$

This is a mixed integer program, and such optimization problems are known to be very hard to solve in general. However, in this case, carefully choosing the blocks of variables is essential. Choosing the blocks of variables carefully also trivializes the underlying constraints. In this particular case, the variables are divided into two blocks corresponding to the $k \times d$ prototype variables in the vectors $\bar{z}_1 \dots \bar{z}_k$ and the $n \times k$ assignment variables $Y = [y_{ij}]$. We alternately minimize over these two blocks of variables, because it provides the best possible decomposition of the problem into smaller subproblems. Note that if the prototype variables are fixed, the resulting assignment problem becomes trivial and one assigns each point to the nearest prototype. On the other hand, if the cluster assignments are fixed, then the objective function can be decomposed into separate objective functions over different clusters. The portion of the objective function O_j contributed by the j th cluster is shown by an underbrace in the optimization formulation above. For each cluster, the relevant optimal solution $\bar{z}_j$ is the mean of the points assigned to that cluster. This result can be shown by setting the gradient of the objective function O_j with respect to each $\bar{z}_j$ to 0:

$$\frac{\partial O_j}{\partial \bar{z}_j} = 2 \sum_{i=1}^n y_{ij} (\bar{X}_i - \bar{z}_j) = 0 \quad \forall j \in \{1 \dots k\} \quad (4.76)$$

The points that do not belong to cluster j drop out in the above condition because $y_{ij} = 0$ for such points. As a result, $\bar{z}_j$ is simply the mean of the points in its cluster. Therefore, we need to alternatively assign points to their closest prototypes, and set the prototypes to the centroids of the clusters defined by the assignment; these are exactly the steps of the well-known k -means algorithm. The centroid computation is a continuous optimization step, whereas cluster assignment is a discrete optimization step (which is greatly simplified by the decomposition approach of coordinate descent).

4.11 Summary

This chapter introduces the basic optimization models in machine learning. We discussed the conditions for optimality, as well as the cases in which a global optimum is guaranteed. Optimization problems in machine learning often have objective functions which can be

separated into components across individual data points. This property enables the use of efficient sampling methods like stochastic gradient descent. Optimization models in machine learning are significantly different from traditional optimization in terms of the need to maximize performance on out-of-sample data rather than on the original optimization problem defined on the training data. Several examples of optimization in machine learning, such as linear regression, support vector machine, and logistic regression were discussed. Generalizations to multiclass models were also discussed. An alternative to stochastic gradient descent is coordinate descent, which can be more efficient in some situations.

4.12 Further Reading

Optimization is a field that has applications in many disciplines, and several books with a generic focus may be found in [10, 15, 16, 22, 99]. The work in [22] is particularly notable in providing a detailed exposition on convex optimization. Some of the books on linear algebra [130] are numerically focused, and provide several details of linear optimization algorithms. Methods for numerical optimization are also discussed in [52]. Some of the basic linear algebra books [122, 123] discuss the basics of optimization. A detailed discussion of linear regression methods may be found in linear algebra, optimization, and machine learning books. We recommend the available machine learning books [1–4, 18, 19, 39, 46, 53, 56, 85, 94, 95] cover various machine learning applications.

Least-squares regression and classification dates back to the Widrow-Hoff algorithm [132] and Tikhonov-Arsenin's seminal work [127]. A detailed discussion of regression analysis may be found in [36]. The Fisher discriminant was proposed by Ronald Fisher [45] in 1936, and it turns out to be a special case of least-squares regression in which the binary response variable is used as the regressand [18]. The support-vector machine is generally credited to Cortes and Vapnik [30], although the primal method for L_2 -loss SVMs was proposed several years earlier by Hinton [60]. This approach repairs the loss function in least-squares classification by keeping only one-half of the quadratic loss curve and setting the remaining to zero to create a smooth version of hinge loss (try this on Figure 4.9(a)). The specific significance of this contribution was lost within the broader literature on neural networks. A number of practical implementations of LIBSVM are available in [27] and those of linear classifiers are available in LIBLINEAR [44]. Detailed discussions of SVMs are provided in [31]. Discussions of numerical optimization techniques for logistic regression are provided in [93]. Coordinate descent is discussed in Hastie *et al.* [56], and more recently in [134].

4.13 Exercises

- Find the saddle points, minima, and the maxima of the following functions:

(a) $F(x) = x^2 - 2x + 2$

(b) $F(x, y) = x^2 - 2x - y^2$

- Suppose that $\bar{y}$ is a d -dimensional vector with very small norm $\epsilon = \|\bar{y}\|_2$. Consider a continuous and differentiable objective function $J(\bar{w})$ with zero gradient and Hessian H at $\bar{w} = \bar{w}_0$. Show that $\bar{y}^T H \bar{y}$ is approximately equal to twice the change in $J(\bar{w})$ by perturbing $\bar{w} = \bar{w}_0$ by ϵ in direction $\bar{y}/\|\bar{y}\|$.
- Suppose that an optimization function $J(\bar{w})$ has a gradient of 0 at $\bar{w} = \bar{w}_0$. Furthermore, the Hessian of $J(\bar{w})$ at $w = \bar{w}_0$ has both positive and negative eigenvalues. Show

how you would use the Hessian to (i) find a vector direction along which infinitesimal movements in either direction from $\bar{w}_0$ decrease $J(\bar{w})$; (ii) find a vector direction along which infinitesimal movements in either direction from $\bar{w}_0$ increase $J(\bar{w})$. Is $\bar{w}_0$ a maximum, minimum, or saddle-point?

4. We know that the maximum of two convex functions is a convex function. Is the minimum of two convex functions convex? Is the intersection of two convex sets convex? If the union of two convex sets convex? Justify your answer in each case.
5. Either prove each statement or give a counterexample: (i) If $f(x)$ and $g(x)$ are convex, then $F(x, y) = f(x) + g(y)$ is convex. (ii) If $f(x)$ and $g(x)$ are convex, then $F(x, y) = f(x) \cdot g(y)$ is convex.
6. **Hinge-loss without margin:** Suppose that we modified the hinge-loss on page 184 by removing the constant value within the maximization function as follows:

$$J = \sum_{i=1}^n \max\{0, (-y_i[\bar{W} \cdot \bar{X}_i^T])\} + \frac{\lambda}{2} \|\bar{W}\|^2$$

This loss function is referred to as the *perceptron criterion*. Derive the stochastic gradient descent updates for this loss function.

7. Compare the perceptron criterion of the previous exercise to the hinge-loss in terms of its sensitivity to the magnitude of $\bar{W}$. State one non-informative weight vector $\bar{W}$, which will always be an optimal solution to the optimization problem of the previous exercise. Use this observation to explain why a perceptron (without suitable modifications) can sometimes provide much poorer solutions with an SVM when the points of the two classes cannot be separated by a linear hyperplane.
8. Consider an unconstrained quadratic program of the form $\bar{w}^T A \bar{w} + \bar{b}^T \bar{w} + c$, where $\bar{w}$ is a d -dimensional vector of optimization variables, and the $d \times d$ matrix A is positive semidefinite. The constant vector $\bar{b}$ is d -dimensional. Show that a global minimum exists for this quadratic program if and only if $\bar{b}$ lies in the column space of A .
9. The text of the book discusses a stochastic gradient descent update of the Weston-Watkins SVM, but not a mini-batch update. Consider a setting in which the mini-batch S contains training pairs of the form $(\bar{X}, c)$, where each $c \in \{1, \dots, k\}$ is the categorical class label. Show that the stochastic gradient-descent step for each separator $\bar{W}_r$ at learning rate α :

$$\bar{W}_r \leftarrow \bar{W}_r(1 - \alpha\lambda) + \alpha \sum_{(\bar{X}, c) \in S, r=c} \bar{X}^T [\sum_{j \neq r} \delta(j, \bar{X})] - \alpha \sum_{(\bar{X}, c) \in S, r \neq c} \bar{X}^T [\delta(r, \bar{X})] \quad (4.77)$$

Here, $\bar{W}_r$ is defined in the same way as the text of the chapter.

10. Consider the following function $f(x, y) = x^2 + 2y^2 + axy$. For what values of a (if any) is the function $f(x, y)$ concave, convex, and indefinite?
11. Consider the bivariate function $f(x, y) = x^3/6 + x^2/2 + y^2/2 + xy$. Define a domain of values of the function, at which it is convex.